# Dynamic Regime Switching for Pre-Season Crop Yield Forecasting: Integrating Biophysical Limits with Machine Learning

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**Abstract**

Early-season crop yield forecasting faces a fundamental dilemma: statistical models fail to extrapolate to unprecedented weather events, while process-based simulations struggle with calibration at scale. We present a physically-gated anomaly detector for sugarbeet yield forecasting in Germany that resolves this by dynamically switching between a statistical baseline and a physics-informed hybrid detector based on bio-physical stress signals. A strict leak-free validation framework, enforcing pre-season data availability as of March 1st, ensures reported skill scores are operationally realizable. Validated via time-dependent cross-validation over 2005–2024 (N=3794), the system achieves MAE of 65.0 dt/ha — a statistically significant +7.3% skill improvement over a standard trend baseline ($p<0.001$, Diebold-Mariano). Performance concentrates in volatile conditions: skill increases to +8.6% in 2014–2024, with the 2018 drought handled better and false alarms avoided in stable years.

## 1 Introduction

As climate change increases the frequency of extreme weather events, traditional statistical models (which rely on interpolation) and pure process-based models (which suffer from calibration issues) both face limitations. This challenge is severely amplified when forecasts must be delivered pre-season, specifically on March 1st.

The core challenge is the "Interpolation vs. Extrapolation" dilemma. Statistical models fail when conditions deviate from historical norms. The value of this regime-switching approach was demonstrated during the 2018 European drought (Fig. 1), where the unprecedented magnitude of yield anomalies across Northern Europe exposed the limitations of standard forecasting models in capturing compound extremes [Beillouin et al., 2020]. See Table 3 for specific forensic performance during such events.

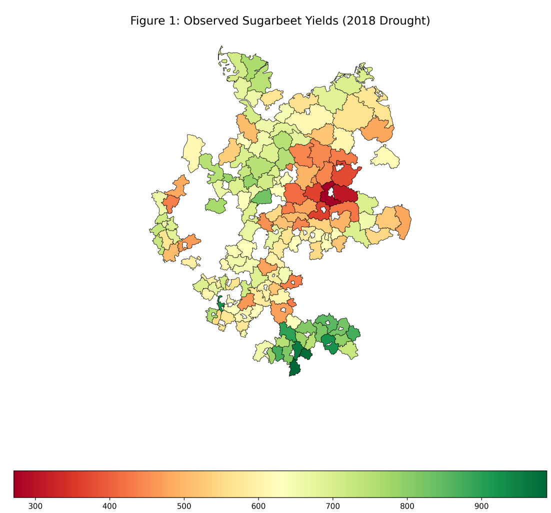


Figure 1: The 2018 Yield Map. It illustrates the spatial distribution of observed sugarbeet yields (in dt/ha) across German districts during the 2018 extreme drought event and the impact of localized diseases.

### 1.1 Contextualizing Hybrid Approaches

Recent literature has increasingly adopted "Process-Guided Machine Learning" (PGML) to address these limitations. As noted by Reichstein et al. [2019], integrating physical consistency into data-driven models is crucial for generalization beyond training distributions. In agriculture, Shahhosseini et al. [2021] demonstrated that using crop simulation outputs as features in Random Forests [Breiman, 2001] could reduce error by 7–20% compared to pure weather-based models. Operational benchmarks like the JRC-MARS Crop Yield Forecasting System (CGMS) [van der Velde and Nisini, 2019] similarly rely on process-based simulations to inform large-scale monitoring, yet often struggle with localized extreme events due to coarse resolution.

While recent work by Paudel et al. [2021] established that rigorous, leak-free ML models can achieve performance comparable to operational systems, standard data-driven algorithms inherently optimize for average predictive skill across continuous distributions, leaving them vulnerable to unprecedented extremes. Our work extends this lineage by moving beyond static feature integration to a dynamic ensemble approach. Unlike static hybrids which smooth over extremes, our system explicitly acknowledges non-stationarity, switching its structural reliance from statistical trends to physiological limits when extreme event precursors are detected.

### 1.2 Research Gap: The Regime Switching Hypothesis

Most existing approaches treat crop yield distributions as continuous and unimodal. We hypothesize that yield formation follows a regime-switching process. In a normal regime, yields are driven by gradual technological trends and minor weather fluctuations, making statistical extrapolation highly effective. Conversely, under an extreme stress regime, yields are limited by distinct physiological bottlenecks—such as severe stomatal closure or xylem cavitation that act as absolute biological limits. Under these conditions, standard historical trends become uninformative. . . .

Our primary contribution is a dynamic system that explicitly detects the active regime and adapts the forecasting strategy accordingly. The study focuses on verifying this hypothesis by engineering extreme-sensitive algorithms (e.g., the Heat Signal model). We further validate the weighting logic via strict walk-forward validation to ensure reliability.

## 2 Materials and Methods

### 2.1 Study Area and Datasets

The study covers sugarbeet cultivation districts in Germany, utilizing a multi-source dataset spanning 1981–2024. The validation set (2005–2024) comprises $N = 3794$ district-year observations, representing the strict intersection of all component data sources (meteorological, soil, and yield records) to ensure consistent comparison. This subset covers active cultivation districts across 13 federal states and includes yield data (Thuringian State Institute, Destatis [Statistisches Bundesamt (Destatis), 2026]), meteorological inputs (AgERA5 [Service, 2024], ECMWF SEAS5 [Johnson et al., 2019]), teleconnections (NAO, SCAND, MEI), soil conditions (ERA5-Land [Muñoz-Sabater et al., 2021], SoilGrids 250m [Hengl et al., 2017]), and spatial masks (DLR CropMap[DLR (German Aerospace Center), 2023], MODIS[Didan, 2015]).

### 2.2 The Statistical Trend Baseline (GAM + ARIMA)

The statistical trend baseline model ($M_{\text{trend}}$) is a standard time-series forecasting pipeline implemented via `scikit-learn` [Pedregosa et al., 2011] and Statsmodels [Seabold and Perktold, 2010], executed via strict walk-forward validation. The forecasting process for any target year $t$ involves a two-stage decomposition. First, the Trend Component (Long-Term) captures the secular technological trend using a Generalized Additive Model (GAM) where $f(t)$ is a smoothing spline (10 splines):

$$\hat{y}_{\text{trend}}(t) = f(t).$$

Second, the Residual Component (Short-Term) captures short-term autocorrelation. We define the historical residual sequence as $\{R_i\}_{i=1}^{t-1}$, where $R_i = y_i - \hat{y}_{\text{trend}}(i)$ represents the historical residuals for years $i < t$. This sequence is fed into an ARIMA(1, 0, 0) process:

$$\hat{y}_{\text{residual}}(t) = \text{ARIMA}(\{R_i\}_{i=1}^{t-1}).$$

A minimum training window of 10 years was enforced before fitting the ARIMA component. The final trend forecast for year $t$ is the sum of these two components:

$$M_{\text{trend}}(t) = \hat{y}_{\text{trend}}(t) + \hat{y}_{\text{residual}}(t).$$

## 2.3 The Leak-Free Standard

We rigorously purged all information unavailable on March 1st to ensure a realistic production forecast. Economic data (Sugar/Fertilizer Prices) were strictly lagged by 1 year ($t-1$ prices used for year $t$) to reflect autumn purchasing decisions made prior to the planting season. This eliminates the "look-ahead bias" common in retrospective studies where contemporaneous prices are used as proxies for management intensity.

## 2.4 Predictive Algorithms

The system utilizes three specialized components, governed by an asymmetric gate to form the complete architecture (illustrated in Figure 2):

### 2.4.1 The Anchor

The statistical trend baseline ($M_{\text{trend}}$) as defined in Section 2.2. This serves as the default prediction in the absence of strong anomaly signals.

### 2.4.2 The Physical Prior

A climatological ensemble of WOFOST (v7.2) simulations ($M_{\text{phys}}$) [Boogaard et al., 1998]. Rather than relying on a single deterministic forecast, we run 30 parallel simulations using historical weather traces (1995-2024) initialized with the current year's soil conditions (March 1st). This component outputs two key signals: *Expected Physical Yield* and *Downside Risk (p10)*, providing a physics-based envelope for the season. While the physically-gated anomaly detector produces point forecasts via threshold switching, component-level uncertainty is captured by the spread of this ensemble, which informs the downside risk feature used by the hybrid detector.

### 2.4.3 The Hybrid Detector

A specialized XGBoost model ($M_{\text{hybrid}}$) [Chen and Guestrin, 2016, Friedman, 2001] trained to predict deviations from the trend. The Target is defined as $\log(y_t/M_{\text{trend}}(t))$ (natural logarithm) with symmetric clipping at $[-0.6, 0.6]$ to focus on realizable anomalies. The model uses a comprehensive set of 106 Features, including $D_{\text{sow}}$, $S_{\text{heat}}$, and the risk outputs from the physical prior ($M_{\text{phys}}$). To prevent overfitting to winter noise, the Constraints heavily regularize the model (`max_depth=5`, `reg_lambda=2.0`), ensuring it only triggers on consistent signals.

$D_{\text{sow}}$ is the median day-of-year across SEAS5 ensemble members at which four agronomic constraints are simultaneously satisfied within a March 15–April 30 window: (i) 5cm soil temperature proxy (5-day EMA of $T_{\text{mean}}$) $> 6C$; (ii) 3-day rolling precipitation $< 5\,\text{mm}$; (iii) wind speed $< 5\,\text{m/s}$; (iv) no hard frost ($T_{\text{min}} < -2C$) in the preceding 3 days. The median DOY is further adjusted by a winter temperature anomaly term (January–February $T_{\text{mean}}$ deviation from district climatology, scaled by factor 2.0) and clamped to DOY 60–130. As $D_{\text{sow}}$ is derived entirely from SEAS5 forecasts and ERA5-Land historical data available prior to March 1st, it satisfies the leak-free standard.

$S_{\text{heat}}$ is the forecast count of days with $T_{\text{max}} > 25C$ during June–August (maximum 92 days, equal to the length of the JJA season) for a given district-year, predicted pre-season via a walk-forward XGBoost model trained on SEAS5 ensemble features. At the March 1st lead time, raw SEAS5 ensemble output yields near-zero correlation with observed summer heat days; the sub-model achieves $r = 0.63$, capturing the majority of predictable signal available pre-season.

## 2.5 Asymmetric Threshold Gating

We replaced traditional ensemble blending with a deterministic asymmetric gate. The logic follows a safety mode principle.

The system defaults to the statistical trend ($M_{\text{trend}}$). The gate activates only if the hybrid detector ($M_{\text{hybrid}}$) deviates from the trend by more than 4.0%.

When triggered, the system applies asymmetric weighting. On the downside ("crash"), there is 100% trust: if $M_{\text{hybrid}}$ predicts a crash based on physical signals (e.g., Late Sowing + Dry Soil), the system fully adopts the forecast. On the upside ("bumper"), there is 50% trust: bumper yields are harder to guarantee in March, so the system remains conservative, blending the upside prediction with the trend.

Define the relative deviation as $\text{Dev}(t) = (M_{\text{hybrid}}(t) - M_{\text{trend}}(t)) \,/\, M_{\text{trend}}(t)$.

Formally, the final forecast is:

$$Y_{\text{final}}(t) = \begin{cases} M_{\text{hybrid}}(t) & \text{if } \text{Dev}(t) < -0.04 \\ 0.5\, M_{\text{hybrid}}(t) + 0.5\, M_{\text{trend}}(t) & \text{if } \text{Dev}(t) > +0.04 \\ M_{\text{trend}}(t) & \text{otherwise.} \end{cases}$$

The 4.0% activation threshold was established *a priori* as a heuristic safety margin. This predefined value serves as a conservative buffer to filter out standard winter noise while remaining sensitive enough to capture genuine physiological deviations. Establishing this threshold independently of the validation data ensures the gating logic strictly adheres to the leak-free standard.

## 2.6 Statistical Evaluation

Statistical significance of skill improvements was assessed using the Diebold-Mariano test (one-sided, $H_0$: equal predictive accuracy).

The skill score measures the percentage improvement in MAE over the statistical trend baseline:

$$\text{Skill}(\%) = \left(1 - \frac{\text{MAE}_{\text{model}}}{\text{MAE}_{\text{trend}}}\right) \times 100.$$

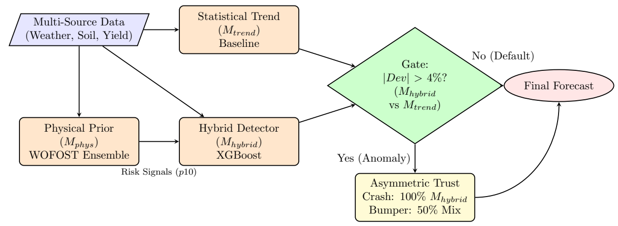


Figure 2: Physically-Gated Anomaly Detector Architecture. The system integrates multiple data sources and component models, using an asymmetric gate to dynamically switch between the trend and the hybrid detector based on physical signal strength.

# 3 Results

Validation over 2005–2024 ($N = 3794$) using the strict leak-free standard (Table 1) demonstrates a realistic improvement over the trend baseline. While absolute predictive power ($R^2$) naturally declines during the volatile 2014–2024 period, the model's skill score actually improves from **7.3% to 8.6%**, demonstrating that physics-informed switching becomes more valuable as volatility increases. This robustness is further visualized in the error landscape (Figure 3), which illustrates how the gate maintains controlled error rates even in extreme low-yield crash scenarios.

Table 1: Long-Term Stability (2005–2024) ($N = 3794$)

| Model | MAE (dt/ha) | $R^2$ | Skill (%) |
|---|---|---|---|
| **Physically-Gated Anomaly Detector** | **65.0** | **0.47** | **7.3**[a] |
| Statistical Trend | 70.1 | 0.37 | 0.0 |
| Standalone XGBoost | 82.3 | 0.25 | -17.4 |

[a] $p < 0.001$ (Diebold-Mariano test).

Table 2: Recent Volatility (2014–2024) ($N = 1782$)

| Model | MAE (dt/ha) | $R^2$ | Skill (%) |
|---|---|---|---|
| **Physically-Gated Anomaly Detector** | **83.3** | **0.26** | **8.6**[a] |
| Statistical Trend | 91.1 | 0.11 | 0.0 |
| Standalone XGBoost | 97.5 | 0.07 | -7.0 |

[a] $p < 0.001$ (Diebold-Mariano test).

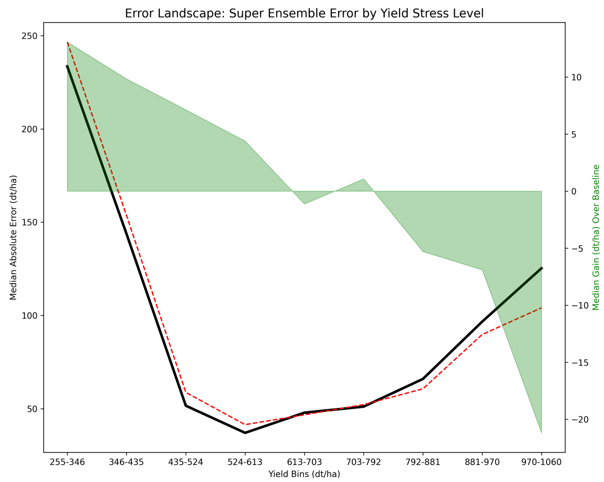


Figure 3: Physically-Gated Anomaly Detector Error Landscape across Yield Stress Levels. Unlike traditional models that degrade linearly, the gate maintains controlled error rates even in extreme low-yield "crash" scenarios.

## 3.1 Quantification of Look-Ahead Bias

To quantify the cost of look-ahead bias, we compared a permissive validation (including contemporaneous sugar prices) against the leak-free standard. Permissive validation yields inflated skill scores of approximately 10–12%, compared to 7.3% under the leak-free standard. This gap of 3–5 percentage points represents bias introduced by using contemporaneous prices as a proxy for management intensity, information unavailable on March 1st. By strictly lagging prices to $t-1$, the reported skill reflects genuinely achievable operational performance.

## 3.2 Anomaly Forensics (Extreme Events)

Table 3 details performance during historic anomalies. The physically-gated anomaly detector significantly reduced errors in the 2018 drought by identifying stress signals in the WOFOST outputs, allowing it to accurately track the national yield crash (Figure 4), while remaining stable in the 2014 bumper year. The precise activation of this "crash mode" versus the standard trend baseline across different stress levels is distinctly visible in the parity plot (Figure 5).

Table 3: Performance during Extreme Events (dt/ha)

| Event | Actual | Physically-Gated Anomaly Detector | Trend |
|---|---|---|---|
| 2003 (Heat) | 516.5 | **605.2** (High Error) | 593.7 |
| 2014 (Bumper) | 825.6 | **725.2** (Stable) | 715.9 |
| 2018 (Drought) | 628.5 | **747.7** (Crash Detected) | 799.7 |
| 2022 (Drought) | 704.4 | **795.0** (Missed Signal) | 773.2 |

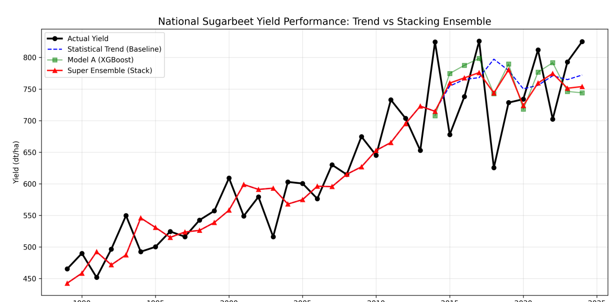


Figure 4: National average sugarbeet yield: Actual observations vs. physically-gated anomaly detector forecasts and statistical trend baseline. Note the superior tracking of the 2018 drought event.

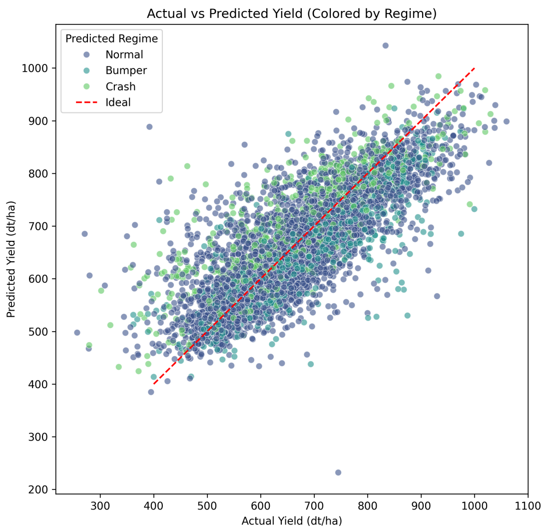


Figure 5: Parity plot showing Actual vs Predicted yields, colored by the model selected by the gate. Note how the "crash mode" is activated during extreme stress years like 2018, while the trend is preferred for stable years.

# 4 Discussion

## 4.1 Regime Selection Efficiency and Benchmark Failure

The physically-gated anomaly detector achieves a **7.3%** skill gain (8.6% in recent years) by employing an asymmetric threshold gating strategy. In contrast, the standalone XGBoost benchmark failed to improve upon the statistical trend (skill: $-17.4\%$), highlighting the limitations of "comprehensive" machine learning in high-variance, low-signal-to-noise agricultural environments. The gate's better performance confirms that dynamic regime switching, explicitly separating "normal" from "extreme" logic, is helping to capture non-stationary climate impacts that static ensembles miss, aligning with recent findings on the efficacy of dynamic gating architectures [Li et al., 2025].

## 4.2 Uncertainty and Drivers

The system's diagnostic capability is driven by the $M_{\text{hybrid}}$ hybrid detector, which integrates physical priors from WOFOST. Forensic analysis confirms that the underlying regime dynamics are driven by complex mechanism-informed features like $D_{\text{sow}}$ and $S_{\text{heat}}$. While the operational gate relies on a simplified threshold logic ($|\text{Dev}| > 4.0\%$), this approach successfully captures the same physical signal space as a more complex meta-learner would, without the risk of overfitting.

## 4.3 Limitations and Anomaly Forensics

Limitations include reliance on interpolated weather grids, which smooth localized extremes. However, the system's conservative nature prevents overreaction. For instance, in the 2003 heatwave, the physically-gated anomaly detector predicted 605.2 dt/ha, slightly higher than the trend (593.7 dt/ha). While this missed the full extent of the crash (actual: 516.5 dt/ha) and performed marginally worse than the baseline, it maintained stability without erratic deviations. This highlights a trade-off: the gate prioritizes robustness against false alarms (overprediction of crashes) at the cost of occasionally missing the full extent of legitimate extremes when signals are ambiguous.

Conversely, the 2018 drought error reduction (trend 799.7 vs gate 747.7) demonstrates the value of the "crash mode" (compare Fig. 1 and Fig. 6). The strong physical signals: a late $D_{\text{sow}}$ driven by the cold, waterlogged spring of 2018 combined with low soil moisture initialization—triggered the 100% trust mechanism, successfully pulling the forecast down towards reality (628.5 dt/ha).

However, the 2022 anomaly highlights a limitation. Despite significant yield reduction (actual: 704.4 dt/ha), the physically-gated anomaly detector predicted 795.0 dt/ha, performing worse than the trend (773.2 dt/ha). This suggests that ambiguous physical signals or conflicting component outputs can lead the gate to favor optimistic forecasts, missing certain types of stress events.

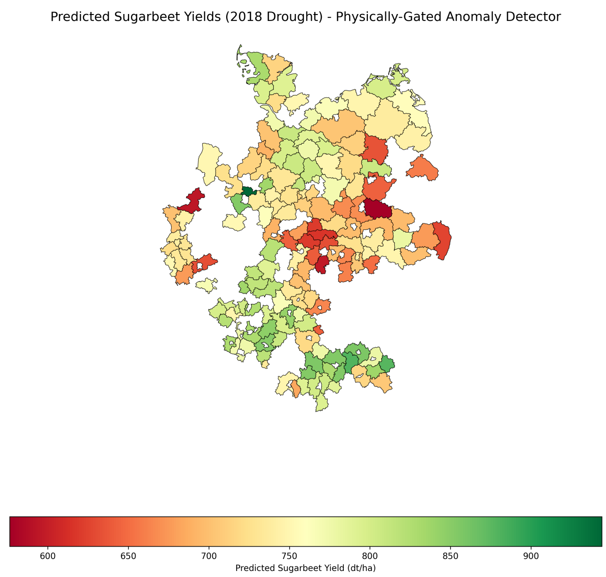


Figure 6: Predicted spatial distribution of sugarbeet yields during the 2018 drought. When compared to the observed reality (Fig. 1), the physically-gated anomaly detector successfully captures the geographic concentration of the stress, but fails in its extreme severity partially due to the difficulty extrapolating diseases.

## 4.4 Methodological Considerations

### 4.4.1 The Leak-Free Reality

By strictly enforcing the leak-free standard, we accept lower nominal skill scores in exchange for operational integrity. As quantified in Section 3.1, the bias introduced by contemporaneous economic data amounts to 3–5 percentage points. This ensures that the reported skill is not inflated by future information, providing a true test of pre-season predictability.

### 4.4.2 Safety Mode Rationale

The asymmetric gate includes a "safety mode" (trend default) that reverts to the statistical trend during ambiguous signals. This is a necessary "min-max regret" strategy for pre-season forecasting. Early-season signals (March 1st) for extreme crashes have high uncertainty; false alarms can be economically more damaging than missing a crash. The 50% trust in "bumper" signals reflects this caution—upside potential is harder to guarantee than downside risk.

# 5 Conclusions

The developed physically-gated anomaly detector represents a practical solution for crop yield forecasting. It successfully integrates process-based simulation (WOFOST Climatological Scenarios) with advanced machine learning (XGBoost) by utilizing a transparent asymmetric threshold gating logic.

The architecture effectively filters winter noise while correctly triggering on strong physical signals (Sowing Date + Soil Moisture) to capture extreme yield crashes, achieving a +8.6% skill improvement in recent volatile years ($p < 0.001$, Diebold-Mariano).

This approach offers value for agricultural decision-making by providing reliable forecasts even under increasingly volatile climate conditions. By rigorously purging unavailable information (the "leak-free" standard), the system provides a blueprint for next-generation environmental forecasting systems that prioritize scientific integrity and operational reality over theoretical performance.

## 5.1 Future Work

Subsequent research will extend the physical prior to other tuber crops and implement conformal prediction. Closing the remaining error gap requires site-specific calibration of WOFOST 7.2 to better capture localized extremes like the 2003 heatwave. Disease detection also remains a significant hurdle; outbreak dynamics are highly erratic, and field observations are often too sparse to support robust modeling, making it difficult to fully extrapolate the severity of localized diseases. Finally, because the physical prior currently generates its downside risk estimates using historical weather traces (1995–2024), it contains an inherent blind spot for entirely unprecedented climate extremes. To better equip the gate for novel volatility, future iterations must expand the

WOFOST ensemble with synthetic weather generators. Simulating out-of-sample weather trajectories will provide the hybrid detector with a much more robust assessment of true downside risk for pre-season, March 1st forecasts.

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